

Smart Academy Management & Learning Platform.

Project Overview

This project is designed to provide educational institutions with a complete digital ecosystem. It manages everything from basic administration and payments to advanced features like live classes, automated exams, and a creative "Talent Hunt" module.

1. User Roles & Access Control

The system operates on a hierarchical role-based access control (RBAC) model:

Admin: Has full control, managing users, academic setups, and viewing data analytics/reports.

Staff/Officer: Focuses on operational tasks like student management, attendance, payment tracking, and publishing notices.

Teacher: Handles academic delivery, including course creation, hosting live classes, creating exams, publishing results, and conducting training sessions.

Student: The end-user who accesses courses, joins live classes, practices exams, views results, manages their profile, and participates in "Talent Hunt" activities.

2. Core Feature Modules

I. Academic & Course Management

Course Structure: Organize courses by class and manage specific subjects.

Content Delivery: Support for various media types, including PDF and video uploads.

Enrollment: Streamlined process for students to enroll in specific courses.

II. Advanced Exam System

Multi-Format Exams: Supports MCQ (with auto-grading), Creative Questions (CQ) with manual marking, and real-time Live Exams.

Smart Features: Includes exam timers, question randomization, automatic result calculation, and integrated ranking systems.

III. Live Class Module

Integration: Utilizing Zoom API (with WebRTC planned for future updates).

Management: Features for scheduling classes, tracking attendance automatically, and providing recorded sessions for later viewing.

IV. Result, Ranking, & Analytics

Automation: Manual marks entry leads to automatic GPA calculation.

Competitive Tracking: Class-wise ranking and a specialized "Talent Hunt" leaderboard to encourage student engagement.

Dashboards: Role-specific views (e.g., Admins see revenue/activity; Teachers see subject-wise performance).

V. Talent Hunt & Creative Work

A unique module for holistic development:

Submissions: Students can submit poetry, stories, and essays for competitions.

Review & Showcase: Teachers review submissions, with an optional public showcase feature.

VI. Financial & Communication Tools

Payment System: Integrated online payments (SSLCommerz/bkash API) for fees and courses, providing automated PDF invoices/receipts.

Messaging: Direct student-teacher messaging and mentor support.

Notifications: Centralized notice board with future plans for Email/SMS alerts.

3. Technical Architecture & Security

Database Core Idea

The database is structured to support multi-tenancy:

Main Tables: Users (IDs, Roles, Passwords), Roles.

Academic/Exam Tables: Courses, Classes, Subjects, Exams, Questions, Answers, Results.

Functional Tables: Live Classes, Submissions (Creative), Payments, Messages.

Security Design

Authentication: Secured via JWT (JSON Web Tokens).

Authorization: Strict Role-Based Access Control (RBAC).

Data Integrity: API security and strict **Data Isolation** to ensure one institution cannot access another's data.

4. Development Roadmap

Phase

Focus

Key Features

Step 1 (MVP)

Core Infrastructure

Login system, Role management, Student/Teacher/Course modules.

Step 2

Academic Depth

Exam system, Result processing, and User Profile management.

Step 3

Interactivity

Live class integration, Payment gateway, and Communication modules.

Step 4

Specialization

Talent Hunt, Training sessions, and the Ranking system.

5. Fully Online Admission: Digitizing the entire entry process.